# 47202 I.F.E. Problems, Lecture 8 2026
# Solutions

**Short questions**

1 Does the electrolyte always have to be an ion conductor (could it be an electron conductor instead)?

Yes (no). The only useful way to close the electrical circuit is to conduct the formed ions. The circuit can also be closed by an electronic conductor, but then the cell is short circuited and no hydrogen production (electrolyzer) or electricity production (fuel cell) will take place.

2 What is the purpose of the catalysts?

A catalyst eases a chemical or an electrochemical process. It takes part in the process, but is not consumed. Without the catalyst, the overvoltage would be so big that the cell would not work at all.

3 What limits the theoretical maximum efficiency of a hydrogen powered fuel cell?

Thermodynamics. The net consumption of gas molecules reduces entropy and therefore part of the energy liberated from the process must be used to give back that entropy. Consequently, the free energy for generating work is less.

4 What limits the practical efficiency of a hydrogen powered fuel cell?

In addition to thermodynamics, overvoltages. Ohmic resistance, mainly in the electrolyte and kinetic resistance in the electrode. Due to these overvoltages, which scale with current, the cell is closest to maximum efficiency at low currents and mass transport limitations do not play a role there.

5 The common types of fuel cells use electrolytes, which conduct either $OH^-$, $H^+$ or $O^{2-}$ (or $CO_3^{2-}$). Can you imagine another electrolyte conducting other species?

In principle, other electrolytes can be developed, but the ions conduced have to be part of the electrode reactions. They need to be produced and consumed.

6 Which electrolyzer types are fully commercial today?

AEC (with diaphragm) and PEMEC.

7 Which type of electrolyzer has typically the highest voltage efficiency?

SOEC

8 Which type of electrolyzer has typically the highest current density?

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PEMEC

9 Why is the alkaline electrolyzer the first commercial and still the most common electrolyzer type while the alkaline fuel cell is not?

The alkaline fuel cell is sensitive to $CO_2$ in the air. If the fuel cell is operated on air, as usual, the electrolyte carbonizes, i.e., forms carbonate. In the electrolyzer, air is not entering the cell, so this is not a problem.

10 The traditional alkaline electrolyzer can be pressurized moderately, but, in contrast to the PEM electrolyzer, only with symmetric pressure. Why is that?

The separator (the diaphragm) is porous. With unbalanced pressure the electrolyte in the pores may be pushed through. Electrolyte will build up at the low-pressure side and eventually, the pores may blow out leading to gas mixing.

11 Does the Sabatier reaction increase or decrease the energy content of the fuel (the generated methane with respect to consumed hydrogen)?

The $\Delta H$ of the process is negative this means that it is exothermal, i.e., that energy is liberated. First law (energy conservation) dictates that the energy of the fuel must be less by the same amount.

**Calculations:**

12 A hydrogen/oxygen fuel cell is operated at 0.7 V. What is the electrical efficiency (voltage efficiency) with respect to the thermo-neutral voltage of 1.48 V?

1.48 V corresponds to 100 % electrical efficiency with respect to the HHV. Since voltage is proportional to the energy delivered by the electrons, 0.7 V corresponds to 0.7 V/1.48 V = 0.47 or 47 %

13 If the cell above produces 100 kW electric, then how much heat does it produce?

If 47 % is 1 kW, then the total energy is 100 kW/0.47 = 213 kW and the fraction that is not converted to work is converted to heat, i.e. 213 kW – 100 kW = 113 kW

14 If an electrolyzer needs 4.2 kWh $Nm^{-3}$ of hydrogen (the $Nm^3$ at 0 °C), what is the higher heating value (HHV) efficiency?

# Solutions

The minimum HHV energy consumption is 3.54 kWh/Nm$^3$ (energy pr. normal cubic meter calculated at 0 °C). This corresponds to 100 % electrical efficiency since only the energy that is contained in the hydrogen is consumed by the system. Efficiency is what you get divided by what you pay:

$$\eta = \frac{3.54\ \text{kWh/Nm}^3}{4.2\ \text{kWh/Nm}^3} = 0.84 = \underline{84\ \%}$$

15 One stack of the electrolyzer plant in Zimbabwe produced 760 Nm$^3$ h$^{-1}$. If we assume 75 % efficiency (HHV), then what is the nominal electrical power? (the Nm$^3$ at 0 °C)

Use Table 4 in the text, assume that water enters as a liquid and use HHV numbers.

At 100 % efficiency (HHV), $P_{100\%}$ = 760 Nm$^3$ h$^{-1}$ · 3.54 kWh Nm$^{-3}$ = <u>2690 kW</u>

At 75 % efficiency (HHV), $P_{75\%}$ = 2690 kW/0.75 = 3587 kW = <u>3.6 MW</u>

16 The enthalpy of combustion of methane to water and $CO_2$ is -890.8 kJ mol$^{-1}$ and the free energy of the process is -818.4 kJ mol$^{-1}$. What is the maximum electrical efficiency and the reversible cell voltage of a fuel cell running on methane and oxygen?

The maximum work (or electrical energy) is the free energy of the reaction. The maximum electrical efficiency is thus the free energy of the reaction divided by the enthalpy of the reaction, $\Delta g/\Delta h$ = -818.4 kJ mol$^{-1}$ / -890.8 kJ mol$^{-1}$ = 0.92 = <u>92%</u>

The reversible cell voltage depends on the free energy. It is -$\Delta g$ divided by the charge transferred. Methane combustion involves 8 electrons (4 for C and 4 for H's for conversion to $CO_2$ and $H_2O$, respectively).

$$E = E = \frac{-\Delta g}{nF} = \frac{818{,}400\ \text{J mol}^{-1}}{8 \cdot 96485\ \text{C mol}^{-1}} = \underline{1.06\ \text{V}}$$

Another way to find the number of electrons is to look at the combustion reaction: unbalanced: $CH_4 + O_2 \rightleftarrows CO_2 + H_2O$

<u>Balanced:</u> $CH_4 + 2O_2 \rightleftarrows CO_2 + 2H_2O$

The oxidant is $O_2$ i.e., 4 oxygen atoms each consuming two electrons ⇒ 8 electrons in total.

17 A fuel cell stack is made from 130 identical single cells each with an electrode area of 64 cm$^2$ (of each electrode) and performance characteristics like in Figure 11 of the text. The stack is operated at 1.5 A cm$^{-2}$. Calculate

a) The total stack voltage

With 1.5 A cm$^2$ the single cell voltage is ca. 0.6 V (read from figure). With series connection and 0.6 V per cell, the total voltage becomes 130 · 0.6 V = 78 V

b) The total stack current

With 1.5 A cm$^2$ and an electrode area of 64 cm, the total cell current is 1.5 A cm$^2$ · 64 cm = 96 A. It is the same current that passes through the entire stack. Total stack current: 96 A

c) The electrical power

The electrical power of the stack, $P_{el}$ is 78 V · 96 A = 7488 W

d) The heat generated assuming water leaves the stack as a liquid at room temperature. The stack is powered by hydrogen.

The total power converted in the stack is the current multiplied by the thermoneutral voltage (times no. of cells) instead of the actual voltage (remember the thermoneutral voltage corresponds to 100 % conversion to electrical energy).

Total power, $P_{tot} = I \cdot E_{TN}$ = 96 A · (130 ·1.48 V) = 18480 W

The heat flow must be $P_{heat} = P_{tot} - P_{el}$ = 18480 W - 7488 W = 10982 W

*Alternatively*:

$$\frac{P_{el}}{P_{tot}} = \frac{E_{\text{cell}}}{E_{\text{TN}}}$$

$$\frac{P_{tot} - P_{heat}}{P_{tot}} = \frac{E_{\text{cell}}}{E_{\text{TN}}}$$

$$P_{heat} = P_{tot}\left(1 - \frac{E_{\text{cell}}}{E_{\text{TN}}}\right) = 18480\ \left(1 - \frac{0.6\ V}{1.48\ V}\right) = 10988\ \text{W}$$

18 A traditional alkaline electrolyzer operates at 0.5 A cm$^{-2}$. With large round cells of 1 meter diameter, and with 500 cells in a stack and with a power demand of 4.5 kWh Nm$^{-3}$ hydrogen (0 °C), how many stacks are needed to match a 3 MW wind turbine?

# Solutions

Strategy: Find the total current, then the charge converted per cell per time. From charge per time, get mole hydrogen per time and then normal cubic meter hydrogen per time. Then calculate the power consumption and compare.

Current: $I = A \cdot i = (50\ cm)^2 \pi \cdot 0.5\ \mathrm{A\ cm^{-2}} =$ 3927 A. This is 3927 C $s^{-1}$ $cell^{-1}$.

The total charge converted is $\dot{Q}$ = 3927 C $s^{-1}$ $cell^{-1}$ X 500 cells = 1,963,500 C $s^{-1}$.

The number of moles hydrogen formed pr. second is $\dot{n} = \frac{\dot{Q}}{2F} = \frac{1963500\ \mathrm{C\ s^{-1}}}{2 \cdot 96485\ \mathrm{C\ mol^{-1}}} =$ 10.18 mol $s^{-1}$

The volume pr. time is $\dot{V} = \frac{\dot{n}RT}{P} = \frac{10.18\ \mathrm{mol\ s^{-1}} \cdot 8.314\ \mathrm{J\ mol^{-1} 273 K^{-1}}}{10^5\ \mathrm{Pa}} =$ 0.231 $m^3$ $s^{-1}$ = 832 $m^3$ $h^{-1}$

The electrical power required to run the electrolyzer is $\dot{W}$ = 4.5 kWh/$m^3$ X 832 $m^3$ $h^{-1}$ = 3744 kW = 3.7 MW

It is a little more than needed for the 3 MW wind turbine. One electrolyzer will do.

Alternative solution:

Find the total current like above: 3927 A.

Find the voltage from the power demand of 4.5 kWh $Nm^{-3}$ hydrogen:

We know that the ideal power demand is 3.54 kWh $Nm^{-3}$ which corresponds to a cell voltage of 1.48 V. In this case the energy demand is higher than idea by $\frac{4.5\ \mathrm{kWh\ Nm^{-3}}}{3.54\ \mathrm{kWh\ Nm^{-3}}} = 1.271$. The cell voltage is also higher by the same amount: $E_{\mathrm{cell}} = 1.48 \cdot 1.271 = 1.88$

Total stack voltage $E_{\mathrm{stack}} = 1.88 \cdot 500 = 941\ \mathrm{V}$

Stack power: $P_{\mathrm{stack}} = 941\ \mathrm{V} \cdot 3927\ \mathrm{A} = 3695307\ \mathrm{W} =$ 3.7 MW

19 The annual production of iridium is less than 10 ton. In 2016 it was 7.1 ton. Assuming 1 mg Ir pr. $cm^2$ and a rated electrical power of 2 W $cm^{-2}$, what would the total power be if the total amount of Ir was used for PEM electrolyzers in 2016?

How big a fraction of the world energy consumption of ca. 20 TW would that be?

Total: 7.1 · $10^9$ mg Ir

$P = \frac{7.1 \cdot 10^9\ \mathrm{mg}\ 2\ \mathrm{W}\ cm^{-2}}{1\ mg\ cm^{-2}} = 14.2 \cdot 10^9\ W =$ 14.2 GW

This is $\frac{14.2}{20 \cdot 10^3} = 0.0007$ or 0.07% of the global energy demand. With lower Ir loading the maximum power can increase correspondingly.

20 An electrolyzer in a hydrogen fuelling station operates in a continuous mode consuming 5 kWh $Nm^{-3}$ hydrogen (use 1 atm and 0 °C for the normal cubic meter).

a) Which overall efficiency does that correspond to with respect to HHV and LHV?

HHV = 285.8 kJ/mol, LHV = 241.8 kJ $mol^{-1}$

5 kWh = 5*3600 kJ = 18000 kJ

1 $Nm^3$: $n = \frac{PV}{RT} = \frac{1.0135 \cdot 10^{-5}\ \text{Pa} \cdot 1\ \text{m}^2}{8.314\ \text{J mol}^{-1}\text{K}^{-1} \cdot 273.15\text{K}} = 44.63\ \text{mol}$

Energy consumption: 18000 kJ/44.63 mol = 403.3 kJ $mol^{-1}$

$\eta_{HHV}$ = 285.8 kJ $mol^{-1}$/ 403.3 kJ $mol^{-1}$ = 0.7087 = 70.9 %

$\eta_{LHV}$ = 241.8 kJ $mol^{-1}$/ 403.3 kJ $mol^{-1}$ = 0.5996 = 60.0 %

Alternative:

Use ideal values from slide: HHV (0 °C, 1 atm) 3.54 kWh $m^{-3}$

LHV (0 °C, 1 atm) 3.00 kWh $m^{-3}$

$\eta_{HHV}$ = 3.54 kWh $m^{-3}$ / 5 kWh $m^{-3}$ = 0.708 = 70.8 %

$\eta_{LHV}$ = 3.00 kWh $m^{-3}$ / 5 kWh $m^{-3}$ = 0.600 = 60.0 %

b) If balance of plant components consume 10 % of the total electrical energy, what is the cell voltage (assuming all cells perform identically)?

$\eta_{HHV} = E_{TN} / E_{cell} \Rightarrow E_{cell} = E_{TN} / \eta_{HHV}$

10 % for BOP, then only 90 % of the input electricity is used in the electrolyzer cells for the process. Thus the voltage efficiency is higher than the system efficeincy:

Voltage efficiency (HHV, the cell alone) = 70.9% /0.90 = 78.78 %

$E_{cell} = E_{TN} / \eta_{HHV}$ = 1.48 V / 78.78 % = 1.88 V

c) How much water is needed for the production of 4 kg (the need of a typical FC vehicle) of hydrogen under these conditions?

# Solutions

$M_{H2}$ = 2 g $mol^{-1}$, $M_{H20}$ = 18 g $mol^{-1}$. Ratio 1:9

4 kg $H_2$ is made from 4*9 kg of water, i.e. 36 kg = 36 L